5.2 Narrative Prediction

The models also do better with a direct prompt than a narrative prompt. GPT-4.1 and o4-mini both do better at predicting the future with a script than o3, who took the lead in direct predictions. Deepseek v3 seemed better at direct prediction than Deepseek r1, but the scores are reversed here. Brier score breakdown by category for narrative questions can be found in Table 16 in Appendix E.

Model	Median Ensemble		Mean Ensemble	
	Score	Std Error	Score	Std Error
GPT-4.1	0.1842	0.0196	0.1726	0.0155
Qwen3-235B-A22B	0.1871	0.0175	0.1801	0.0132
GPT-4o	0.1940	0.0148	0.1991	0.0126
o4-mini	0.1977	0.0208	0.1976	0.0185
o3	0.1985	0.0196	0.1982	0.0189
o3-pro	0.2033	0.020	0.1933	0.0178
Qwen3-32B	0.2178	0.0195	0.2218	0.0162
Claude-3.6-Sonnet	0.2345	0.0184	0.2267	0.0149
Claude-3.5-Sonnet	0.2415	0.0160	0.2278	0.0111
Deepseek-r1	0.2454	0.0217	0.2371	0.0193
Deepseek-v3	0.2689	0.0130	0.2633	0.0119
GPT-4o-mini	0.3047	0.0116	0.2942	0.0109

Table 5: Median and mean Brier scores for narrative prediction

5.3 Hold-out Set Results

On the hold-out set, the models do similarly to the larger question set. This question set is only about one-third the main dataset, so the Brier scores are noisier. I also run o3-pro on this smaller dataset, and it gets a Brier score of 0.1307. o3 does slightly worse than previously (going from 0.1352 to 0.1307), but GPT-4.1 does exactly the same as in the previous set, Claude 3.6 Sonnet does slightly better, and GPT-4o does much better. Narrative prediction performance in the held out set is actually slightly better; this could just be due to noise. The underlying distribution of questions is roughly the same. Categorization of scores on the hold-out set are in Appendix E.

Model	Median Ensemble		Mean Ensemble	
	Score	Std Error	Score	Std Error
o3-pro	0.1307	0.0148	0.1303	0.0146
o3	0.1375	0.0156	0.1392	0.0152
GPT-4.1	0.1575	0.0209	0.1538	0.0196
o4-mini	0.1626	0.0206	0.1610	0.0196
GPT-4o	0.1765	0.0178	0.1739	0.0165
Claude-3.6-Sonnet	0.1801	0.0180	0.1762	0.0163
Deepseek-v3	0.1840	0.0168	0.1825	0.0159
Claude-3.5-Sonnet	0.1891	0.0176	0.1878	0.0166
Qwen3-235B-A22B	0.1929	0.0183	0.1973	0.0178
Deepseek-r1	0.1994	0.0188	0.2008	0.0177
Qwen3-32B	0.2145	0.0216	0.2103	0.0201
GPT-4o-mini	0.2781	0.0219	0.2701	0.0203

Table 6: Median and mean Brier scores for direct prediction on held-out set

The numbers for narrative are worse than direct prediction, but they generally are better than narrative predictions on the previous dataset. Claude 3.6 Sonnet does a lot worse, but o3, GPT-4.1, o3-pro, and Deepseek r1 all do better.

Model	Median Ensemble		Mean Ensemble	
	Score	Std Error	Score	Std Error
o3	0.1544	0.0162	0.1559	0.0148
GPT-4.1	0.1693	0.0219	0.1641	0.0199
o3-pro	0.1723	0.0173	0.1684	0.0160
Qwen3-2335B-A22B	0.1898	0.0209	0.1845	0.0172
Qwen3-32B	0.1792	0.0154	0.1810	0.0142
o4-mini	0.1981	0.0221	0.1827	0.0189
GPT-4o-2024-08-06	0.2127	0.0165	0.2114	0.0137
Claude-3.5-Sonnet	0.2213	0.0165	0.2090	0.0141
Deepseek r1	0.2273	0.0218	0.2308	0.0231
Deepseek v3	0.2676	0.0194	0.2591	0.0173
Claude-3.6-Sonnet	0.2687	0.0235	0.2560	0.0204
GPT-4o-mini	0.3073	0.0173	0.2961	0.0157

Table 7: Median and mean Brier scores for narrative prediction on held-out set

5.4 Expert Forecasters

Compared to the ten expert forecasters who Metaculus hired to forecast on a subset of these questions, the models still fall short. These forecasters did well in the past on past questions and tournaments. Individual models may be roughly equivalent to human crowd, but the crowd score of expert forecasters is still better than the models. Compared to model performance, expert forecasters still significantly outperform the bots with a Brier score of 0.0225, far lower than o3 which got a score of 0.1352.

Metric	Value
Number of Questions	157
Mean Brier Score	0.1573
Median Brier Score	0.0225
Standard Error	0.0189

Table 8: Expert forecasters' performance on main dataset

Experts usually outperform regular individuals consistently (Mellers et al., 2015). Each update they make to their predictions are typically smaller than regular human forecasters, but they alsoupdate their forecasts more frequently (Karvetski, 2021). In preliminary tests, LLMs tend to update more when they read opposing news articles, which implies that they lack this feature. Experts also tend to be more sensitive to the scope of the question, and make more granular forecasts. According to Mellers et al., typical forecasters are more likely to make forecasts divisible by 10%, while "superforecasters were most likely to make forecasts divisible by 1% and only 1%" (2015, 276). Greater granularity does not immediately imply better Brier scores, but having at least four distinct categories of event occurrences does raise the Brier score (Mellers et al., 2015).

On the hold-out set, the expert forecasters do quite well too. The number of questions is much smaller, so the Brier score is noisier, but they still do much better than o3-pro.

Metric	Value
Number of Questions	41
Mean Brier Score	0.1222
Median Brier Score	0.0196
Standard Error	0.0309

Table 9: Superforecaster performance on held-out dataset

On the 152 question dataset broken down by category, expert forecasters also do slightly better on politics and governance than economics and business. Only politics and economics categories have

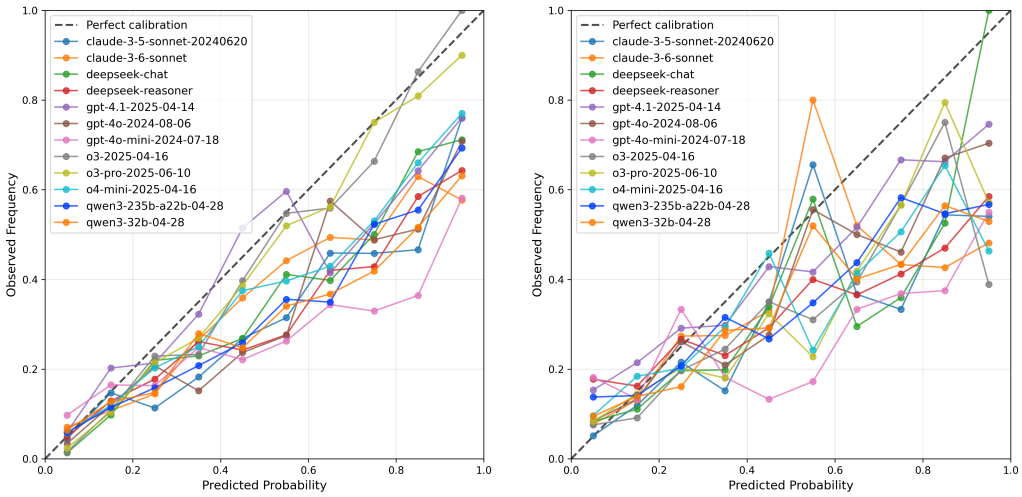
more than 30 questions (see Table 14); the rest have fewer than 20 questions, so the Brier scores are extremely noisy.

Model	Arts & Rec	Economics	Environ.	Healthcare	Politics	Sci & Tech	Sports
Experts	0.0784	0.0307	0.0625	0.0049	0.0169	0.1625	0.0400

Table 10: Superforecaster Median Brier Scores by Category

5.5 Calibration Plots

Calibration plots also show that models are typically overconfident, especially for events that models think are likely to happen. They seem decently well-calibrated for events than have a less than 10% chance of happening, but Qwen-32B for example would predict close to certainty for an event that has an observed frequency of 50%. GPT-4.1 stays quite close to the perfect calibration curve until events that it thinks has a 70% likelihood of occurring, where its accuracy drastically drops.



(a) Calibration plot for direct prediction

(b) Calibration plot for narrative prediction

With a narrative prompt, both Claude Sonnet models are extremely underconfident and predict something has a likelihood of happening at a 50% probability when predicting on events that actually have an 80% likelihood of happening. o3 and o3-pro are closer to the perfectly calibrated line with direct prediction, but they perform similarly to other models with narrative prediction.

6 Conclusion

Models have improved significantly over the past year, and they have surpassed the level of a human crowd, but not a group of experts. o3 achieves a Brier score of 0.1352, outperforming the crowd baseline of 0.149 but underperforming the 0.121 reported in Halawi et al. (2024) and Karger et al. (2025) respectively. This implies that there may be some value in adding LLMs to prediction markets to increase the liquidity, but they still do not perform the same function as a group of experts.

Additionally, models do not perform equally well in all categories. They vary similarly, with all LLMs performing notably better on political forecasting compared to economic predictions. This pattern, consistent across both frontier and older models, suggests that finance and economics outcomes perhaps require a better theory of the world, or maybe they are just stochastic in the short-term. Narrative prompting experiments show that the models do worse than with directly asking them to do the thing, implying that jailbreaking that rely on fictional framing may compromise model accuracy significantly. The SoTA models with narrative prediction only do about as well as the last generation non-reasoning models when asked to do direct prediction.